

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412766
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Goto; Shuhei et al.

Substrate processing apparatus and substrate processing method

Abstract

A substrate processing apparatus **1** includes a controller **61** configured to perform a first recipe and a second recipe in parallel. Each of the first recipe and the second recipe includes a first transfer processing of transferring a substrate from a transit unit **14** to a liquid processing unit **17**, a liquid processing, a second transfer processing of transferring the substrate from the liquid processing unit **17** to a drying unit **18** and a drying processing. The controller **61** determines, while the first recipe on a first substrate in substrates is being performed, when the second recipe on a second substrate in the substrates is started, a start timing of the first transfer processing of the second substrate such that a time period of the second transfer processing of the first substrate does not overlap with a time period of the second transfer processing of the second substrate.

Inventors:	Goto; Shuhei (Koshi, JP), Kaneko; Tomohiro (Kumamoto, JP), Matsumoto; Takeshi (Koshi, JP)
Applicant:	Tokyo Electron Limited (Tokyo, JP)
Family ID:	76221610
Assignee:	TOKYO ELECTRON LIMITED (Tokyo, JP)
Appl. No.:	17/756853
Filed (or PCT Filed):	November 27, 2020
PCT No.:	PCT/JP2020/044378
PCT Pub. No.:	WO2021/112022
PCT Pub. Date:	June 10, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230008278 A1	Jan. 12, 2023

Foreign Application Priority Data

JP 2019-220954 Dec. 06, 2019

Publication Classification

Int. Cl.: H01L21/677 (20060101); B08B3/02 (20060101); B08B13/00 (20060101); F26B5/00 (20060101); H01L21/02 (20060101)

U.S. Cl.:

CPC H01L21/67703 (20130101); B08B3/022 (20130101); B08B13/00 (20130101); F26B5/005 (20130101); H01L21/02057 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5801945	12/1997	Comer	N/A	N/A
2007/0003842	12/2006	Wang et al.	N/A	N/A
2019/0139791	12/2018	Kyose	N/A	N/A
2021/0043471	12/2020	Nakata	N/A	H01L 21/67253

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109768001	12/2018	CN	N/A
2006-190894	12/2005	JP	N/A
2013-012538	12/2012	JP	N/A
2014-022546	12/2013	JP	N/A
2014-033180	12/2013	JP	N/A
10-2019-0038346	12/2018	KR	N/A

OTHER PUBLICATIONS

International Search Report for PCT/JP2020/044378 dated Feb. 9, 2021. cited by applicant

Primary Examiner: Salerno; Sarah K

Attorney, Agent or Firm: Chrisman Gallo Tochtrop LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This Application is a U.S. national phase application under 35 U.S.C. § 371 of PCT Application

No. PCT/JP2020/044378 filed on Nov. 27, 2020, which claims the benefit of Japanese Patent Application No. 2019-220954 filed on Dec. 6, 2019, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The various aspects and exemplary embodiments described herein pertain generally to a substrate processing apparatus and a substrate processing method.

BACKGROUND

(3) Conventionally, there is known a substrate processing apparatus which forms an anti-drying liquid film on a surface of a semiconductor wafer (hereinafter, referred to as a “wafer”) as a substrate and performs a drying processing by bringing the wafer having the liquid film formed thereon into contact with a processing fluid in a supercritical state. Patent Document 1: Japanese Patent Laid-open Publication No. 2013-012538

SUMMARY

(4) In one exemplary embodiment, a substrate processing apparatus includes a transit unit; multiple liquid processing units; multiple drying units; a transfer device and a controller. The transit unit is configured to accommodate therein multiple substrates. The multiple liquid processing units are configured to form a liquid film on a surface of the substrate. The multiple drying units are configured to dry the substrate by bringing the substrate having the liquid film formed on the surface thereof into contact with a supercritical fluid. The transfer device is configured to transfer the substrate. The controller is configured to control the transit unit, the multiple liquid processing units, the multiple drying units and the transfer device to perform two recipes, which includes a first recipe and a second recipe, in parallel, each of the first recipe and the second recipe including a first transfer processing of transferring the substrate from the transit unit to the liquid processing unit, a liquid processing performed by the liquid processing unit, a second transfer processing of transferring the substrate from the liquid processing unit to the drying unit and a drying processing performed by the drying unit, the first recipe and the second recipe having different processing times of at least the liquid processing and the drying processing. Further, the controller determines, while the first recipe on a first substrate in the multiple substrates is being performed, when the second recipe on a second substrate in the multiple substrates is started, a start timing of the first transfer processing of the second substrate such that a time period of the second transfer processing of the first substrate does not overlap with a time period of the second transfer processing of the second substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross-sectional view of a substrate processing system according to an exemplary embodiment, when viewed from the top.
- (2) FIG. 2 is a schematic cross-sectional view of the substrate processing system according to the exemplary embodiment, when viewed from the lateral side.
- (3) FIG. 3 is a side view showing a configuration of a transfer device according to the exemplary embodiment.
- (4) FIG. 4 illustrates a configuration example of a liquid processing unit.
- (5) FIG. 5 illustrates a configuration example of a drying unit.
- (6) FIG. 6 is a flowchart illustrating a sequence of a series of substrate processings performed in the substrate processing system according to the exemplary embodiment.
- (7) FIG. 7 is a timing chart for explaining a subsequent recipe start timing adjustment processing according to the exemplary embodiment.
- (8) FIG. 8 is a timing chart for explaining an example of a subsequent recipe start timing

adjustment processing in consideration of a transfer prohibition period.

(9) FIG. 9 is a timing chart for explaining an example of a method of determining a start timing of a first transfer processing in the same recipe.

(10) FIG. 10 is a timing chart for explaining an example of a method of determining a priority when two recipes can be started.

(11) FIG. 11 illustrates an example of transfer time information.

(12) FIG. 12 is a flowchart illustrating an example of a sequence of a third transfer processing.

DETAILED DESCRIPTION

(13) Hereinafter, embodiments (hereinafter, referred to as “exemplary embodiments”) of a substrate processing apparatus and a substrate processing method according to the present disclosure will be described in detail with reference to the accompanying drawings. Further, the substrate processing apparatus and the substrate processing method of the present disclosure are not limited to the following exemplary embodiments. Furthermore, the exemplary embodiments can be appropriately combined as long as processing contents are not contradictory to each other. Also, in each of the embodiments described below, same parts will be assigned same reference numerals, and redundant description will be omitted.

(14) In order to ease understanding of the following description, the X-axis direction, Y-axis direction and Z-axis direction which are orthogonal to each other are defined, and an orthogonal coordinate system in which the positive Z-axis direction is regarded as a vertically upward direction may be used in the following drawings. Further, a rotational direction around a vertical axis may be referred to as “ θ direction”.

(15) Further, in the following exemplary embodiments, expressions such as “constant,” “perpendicular,” “vertical” and “parallel” may be used. These expressions, however, do not imply strictly “constant,” “perpendicular,” “vertical” and “parallel”. That is, these expressions allow some tolerable errors in, for example, manufacturing accuracy, installation accuracy, or the like.

(16) Conventionally, there is known a substrate processing apparatus which forms an anti-drying liquid film on a surface of a semiconductor wafer (hereinafter, referred to as a “wafer”) as a substrate and performs a drying processing by bringing the wafer having the liquid film formed thereon into contact with a processing fluid in a supercritical state.

(17) However, if another wafer is being transferred at a timing when the wafer having the liquid film formed on its surface is transferred into a subsequent drying unit, the wafer having the liquid film formed thereon is required to stand by until the transfer of the another wafer is completed.

(18) If a state of the liquid film on the surface of the wafer is changed as the liquid film is dried or the like during the standby period, a problem such as a collapse of a pattern formed on the wafer may occur in a subsequent drying processing, which may cause a reduction in yield of wafers.

(19) In particular, the overlap of wafer transfer timings described above is likely to occur when a plurality of recipes having different processing times is performed in parallel.

(20) Therefore, even when a plurality of recipes having different processing times is performed in parallel, it is expected to suppress non-uniformity in transfer time of the wafer having the liquid film on its surface.

(21) <Configuration of Substrate Processing System>

(22) First, a configuration of a substrate processing system 1 (an example of a substrate processing apparatus) according to an exemplary embodiment will be described with reference to FIG. 1 and FIG. 2. FIG. 1 is a schematic cross-sectional view of the substrate processing system 1 according to the exemplary embodiment, when viewed from the top. Also, FIG. 2 is a schematic cross-sectional view of the substrate processing system 1 according to the exemplary embodiment, when viewed from the lateral side.

(23) As shown in FIG. 1, the substrate processing system 1 includes a carry-in/out station 2 and a processing station 3. The carry-in/out station 2 and the processing station 3 are provided adjacent to each other.

(24) The carry-in/out station **2** is provided with a carrier placing section **11** and a transfer section **12**. In the carrier placing section **11**, a plurality of carriers **C** each accommodating therein a plurality of semiconductor wafers **W** (hereinafter, referred to as “wafers **W**”) horizontally is placed.

(25) The substrate processing system **1** according to the exemplary embodiment is configured to perform substrate processings on two types of wafers **W** (hereinafter, referred to as “wafer **WA**” and “wafer **WB**”) with different recipes, respectively. Therefore, the carrier placing section **11** is equipped with a plurality of (herein, two) carriers **CA** each accommodating therein a plurality of wafers **WA** and a plurality of (herein, two) carriers **CB** each accommodating therein a plurality of wafers **WB**.

(26) It is assumed that one of the wafer **WA** and the wafer **WB** is a wafer **W** including a metal on its surface (pattern formation surface) and the other is a wafer **W** not including a metal on its surface. The wafer **WA** and the wafer **WB** just need to be wafers **W** having different processing times of at least a liquid processing and a drying processing, which will be described later, included in a recipe. That is, the term “different types” does not only refer to a difference in surface composition, but also includes a case where the processing times of the liquid processing and the drying processing are different even if the surface composition is the same (i.e., even if the wafers are of the same type).

(27) The transfer section **12** is provided adjacent to the carrier placing section **11**. In the transfer section **12**, a transfer device **13** and a transit unit **14** are placed. As shown in FIG. 2, the transit unit **14** includes a transit unit **14A** capable of accommodating therein the plurality of wafers **WA** in multiple levels and a transit unit **14B** capable of accommodating therein the plurality of wafers **WB** in multiple levels. The transit unit **14A** and the transit unit **14B** are arranged along a vertical direction (**Z**-axis direction).

(28) The transfer device **13** is provided with a wafer holding mechanism configured to hold the wafer **W**. The transfer device **13** is movable horizontally and vertically and pivotable around a vertical axis, and transfers the wafer **W** between the carrier **C** and the transit unit **14** by using the wafer holding mechanism.

(29) Specifically, as shown in FIG. 2, the transfer device **13** includes a first holder **131A** configured to hold the wafer **WA** and a second holder **131B** configured to hold the wafer **WB**. The transfer device **13** may move the first holder **131A** and the second holder **131B** independently of each other in a horizontal direction.

(30) The processing station **3** is provided adjacent to the transfer section **12**. The processing station **3** is provided with a transfer block **4** and a plurality of processing blocks **5**.

(31) The transfer block **4** is provided with a transfer area **15** and a transfer device **16**. The transfer area **15** is, for example, a rectangular parallelepiped region extending along an arrangement direction (**X**-axis direction) of the carry-in/out station **2** and the processing station **3**. The transfer device **16** is placed in the transfer area **15**.

(32) The transfer device **16** is provided with a wafer holding mechanism configured to hold the wafer **W**. The transfer device **16** is movable horizontally and vertically and pivotable around a vertical axis. The transfer device **16** is configured to transfer the wafer **W** between the transit unit **14** and the plurality of processing blocks **5** by using the wafer holding mechanism. Hereinafter, a configuration of the transfer device **16** will be described with reference to FIG. 3. FIG. 3 is a side view showing the configuration of the transfer device **16** according to the exemplary embodiment.

(33) As shown in FIG. 3, the transfer device **16** according to the exemplary embodiment is equipped with a first holder **161A**, a second holder **161B**, a first reciprocating mechanism **162A**, a second reciprocating mechanism **162B**, an elevating mechanism **163** and a horizontal moving mechanism **164**.

(34) The first holder **161A** is configured to hold the wafer **WA**. The second holder **161B** is placed, for example, under the first holder **161A** and configured to hold the wafer **WB**. Each of the first holder **161A** and the second holder **161B** is provided with, for example, a flat plate-shaped base

portion having a bifurcated fork shape whose lateral width is smaller than a diameter of the wafer W and a plurality of supporting members provided on a surface of the base portion. Each of the first holder **161A** and the second holder **161B** supports the wafer W from below by using the plurality of supporting members to horizontally hold the wafer W.

(35) The first reciprocating mechanism **162A** is configured to move the first holder **161A** back and forth along the horizontal direction, more particularly, along the Y-axis direction orthogonal to an extension direction of the transfer area **15**. The second reciprocating mechanism **162B** is configured to move the second holder **161B** back and forth along the Y-axis direction.

(36) The elevating mechanism **163** is configured to move the first reciprocating mechanism **162A** and the second reciprocating mechanism **162B** along the vertical direction to move the first holder **161A** and the second holder **161B** up and down. The horizontal moving mechanism **164** is configured to move the elevating mechanism **163** along the X-axis direction to move the first holder **161A** and the second holder **161B** in the extension direction of the transfer area **15**.

(37) The plurality of processing blocks **5** is placed on one side of the transfer area **15** to be adjacent to the transfer area **15**. Specifically, as shown in FIG. **1** and FIG. **2**, the plurality of processing blocks **5** includes a plurality of processing blocks **5A** configured to process the wafer WA and a plurality of processing blocks **5B** configured to process the wafer WB. The processing blocks **5A** are placed on one side (positive Y-axis direction side) of the transfer area **15** in a direction (Y-axis direction) orthogonal to the arrangement direction (X-axis direction) of the carry-in/out station **2** and the processing station **3**, and the processing blocks **5B** are placed on the other side (negative Y-axis direction side). Also, as shown in FIG. **2**, the plurality of processing blocks **5A** is disposed in multiple levels (herein, three levels) along the vertical direction. Likewise, the plurality of processing blocks **5B** is disposed in multiple levels (herein, three levels) along the vertical direction.

(38) The transfer of the wafer W between the processing block **5** in each level and the transit unit **14** is performed by the single transfer device **16** provided in the transfer block **4**.

(39) Each processing block **5A** is equipped with a liquid processing unit **17** and a drying unit **18**. Likewise, each processing block **5B** is equipped with a liquid processing unit **17** and a drying unit **18**.

(40) The liquid processing unit **17** is configured to perform a cleaning processing of cleaning a top surface which is a pattern formation surface of the wafer W and then a liquid film forming processing of forming a liquid film on the top surface of the wafer W after being subjected to the cleaning processing. The cleaning processing and the liquid film forming processing are examples of the liquid processing. A configuration of the liquid processing unit **17** will be described later.

(41) The drying unit **18** is configured to perform a drying processing on the wafer W after being subjected to the liquid film forming processing. Specifically, the drying unit **18** is configured to dry the wafer W after being subjected to the liquid film forming processing by bringing this wafer W into contact with a processing fluid in a supercritical state (hereinafter, sometimes referred to as "supercritical fluid"). A configuration of the drying unit **18** will be described later.

(42) Further, although not illustrated in FIG. **1** and FIG. **2**, the substrate processing system **1** may be further equipped with a supply unit configured to supply the processing fluid to the drying unit **18**. The supply unit is equipped with a supply device group including a flowmeter, a flow rate controller, a back pressure valve, a heater and the like, and a housing accommodating therein the supply device group. In the present exemplary embodiment, the supply unit supplies the drying unit **18** with CO₂ as the processing fluid.

(43) The liquid processing unit **17** and the drying unit **18** are arranged along the transfer area **15** (i.e., along the X-axis direction). In the liquid processing unit **17** and the drying unit **18**, the liquid processing unit **17** is placed at a position close to the carry-in/out station **2** and the drying unit **18** is placed at a position far from the carry-in/out station **2**.

(44) As described above, each processing block **5** includes one liquid processing unit **17** and one

drying unit **18**. That is, the same numbers of liquid processing units **17** and the drying units **18** are provided in the substrate processing system **1**.

(45) As shown in FIG. **1**, the substrate processing system **1** is equipped with a control device **6**. The control device **6** is, for example, a computer and includes a controller **61** and a storage **62**.

(46) The controller **61** includes a microcomputer having a CPU (Central Processing Unit), a ROM (Read Only Memory), a RAM (Random Access Memory), an input/output port and the like, and various circuits. The CPU of the microcomputer reads out and executes a program stored in the ROM and thus implements controls over the transfer devices **13** and **16**, the liquid processing unit **17**, the drying unit **18** and the like.

(47) Further, this program is stored in a computer-readable recording medium and may be installed to the storage **62** of the control device **6** from this recording medium. The computer-readable recording medium may be, for example, a hard disk (HD), a flexible disk (FD), a compact disk (CD), a magnet optical disk (MO) or a memory card.

(48) The storage **62** may be, for example, a semiconductor memory device such as a RAM or a flash memory, or a storage device such as a hard disk or an optical disk.

(49) <Configuration of Liquid Processing Unit>

(50) Hereinafter, the configuration of the liquid processing unit **17** will be described with reference to FIG. **4**. FIG. **4** illustrates a configuration example of the liquid processing unit **17**. The liquid processing unit **17** is configured as a single-wafer cleaning device configured to clean the wafers **W** one by one by, for example, spin cleaning.

(51) As shown in FIG. **4**, the liquid processing unit **17** holds the wafer **W** substantially horizontally with a wafer holding mechanism **25** provided within an outer chamber **23** forming therein a processing space and rotates the wafer **W** by rotating the wafer holding mechanism **25** around a vertical axis.

(52) Further, in the liquid processing unit **17**, a nozzle arm **26** advances above the wafer **W** being rotated and, the cleaning processing on the top surface of the wafer **W** is performed by supplying a chemical liquid and a rinse liquid in a predetermined sequence from a chemical liquid nozzle **26a** provided at a tip end of the nozzle arm **26**.

(53) Furthermore, in the liquid processing unit **17**, a chemical liquid supply path **25a** is formed within the wafer holding mechanism **25**. A bottom surface of the wafer **W** is also cleaned by the chemical liquid and the rinse liquid supplied through this chemical liquid supply path **25a**.

(54) The cleaning processing may include, for example, a removal of particle or organic contaminant with the chemical liquid, a removal of oxide, a rinse cleaning with the rinse liquid (deionized water: hereinafter, referred to as “DIW”) and the like. The liquid processing unit **17** configured to process the wafer **WA** and the liquid processing unit **17** configured to process the wafer **WB** are different from each other in, for example, the type of the used chemical liquid or the supply time of the chemical liquid.

(55) The above-described various types of chemical liquids are received by the outer chamber **23** and an inner cup **24** disposed within the outer chamber **23**, and then drained out from a drain port **23a** provided at a bottom portion of the outer chamber **23** and a drain port **24a** provided at a bottom portion of the inner cup **24**. Further, the atmosphere within the outer chamber **23** is exhausted through an exhaust port **23b** provided at the bottom portion of the outer chamber **23**.

(56) The liquid film forming processing is performed after the rinse processing in the cleaning processing. Specifically, the liquid processing unit **17** supplies IPA in a liquid state (hereinafter, sometimes referred to as “IPA liquid”) on the top surface and the bottom surface of the wafer **W** while the wafer holding mechanism **25** is being rotated. Accordingly, the DIW remaining on the both surfaces of the wafer **W** is replaced by the IPA. Thereafter, the liquid processing unit **17** gradually stops the rotation of the wafer holding mechanism **25**.

(57) The wafer **W** after being subjected to the liquid film forming processing is delivered to the transfer device **16** by a non-illustrated delivery mechanism provided in the wafer holding

mechanism **25** in a state in which a liquid film of the IPA liquid is formed on the top surface of the wafer **W**, and then, is carried out of the liquid processing unit **17**.

(58) The liquid film formed on the wafer **W** suppresses the occurrence of the pattern collapse caused by evaporation (vaporization) of the liquid on the top surface of the wafer **W** during a transfer of the wafer **W** into the drying unit **18** from the liquid processing unit **17** or during a carry-in operation of the wafer **W** into the drying unit **18**. A processing time of the liquid film forming processing may be different between the liquid processing unit **17** configured to process the wafer **WA** and the liquid processing unit **17** configured to process the wafer **WB**.

(59) <Configuration of Drying Unit>

(60) Hereinafter, a configuration of the drying unit **18** will be described with reference to FIG. 5. FIG. 5 illustrates a configuration example of the drying unit **18**.

(61) As shown in FIG. 5, the drying unit **18** includes a main body **31**, a holding plate **32** and a lid member **33**. The housing-shaped main body **31** is provided with an opening **34** through which the wafer **W** is carried in and out. The holding plate **32** is configured to horizontally hold the wafer **W** as a processing target. The lid member **33** is configured to support the holding plate **32** and hermetically seal the opening **34** when the wafer **W** is carried into the main body **31**.

(62) The main body **31** is a container in which a processing space capable of accommodating therein the wafer **W** having a diameter of, e.g., 300 mm is formed, and is provided with supply ports **35** and **36** and a discharge port **37** in the walls thereof. The supply ports **35** and **36** and the discharge port **37** are respectively connected to supply paths and a discharge path through which the supercritical fluid is circulated into the drying unit **18**.

(63) The supply port **35** is connected to a side surface opposite to the opening **34** in the housing-shaped main body **31**. Further, the supply port **36** is connected to a bottom surface of the main body **31**. The discharge port **37** is connected to a portion of the main body **31** under the opening **34**. Although the two supply ports **35** and **36** and the single discharge port **37** are illustrated in FIG. 4, the number of supply ports **35** and **36** and the number of discharge port **37** are not particularly limited.

(64) Further, fluid supply headers **38** and **39** and a fluid discharge header **40** are provided within the main body **31**. Further, a plurality of supply openings is formed in the fluid supply headers **38** and **39** in a longitudinal direction of the fluid supply headers **38** and **39**, and a plurality of discharge openings is formed in the fluid discharge header **40** in a longitudinal direction of the fluid discharge header **40**.

(65) The fluid supply header **38** is connected to the supply port **35** and provided adjacent to the side surface opposite to the opening **34** in the housing-shaped main body **31**. Further, the plurality of supply openings formed in the fluid supply header **38** is directed toward the opening **34** side.

(66) The fluid supply header **39** is connected to the supply port **36** and provided at a central portion of the bottom surface within the housing-shaped main body **31**. Further, the plurality of supply openings formed in the fluid supply header **39** faces upwards.

(67) The fluid discharge header **40** is connected to the discharge port **37** and provided adjacent to the side surface on the opening **34** side within the housing-shaped main body **31** and below the opening **34**. Further, the plurality of discharge openings formed in the fluid discharge header **40** faces upwards.

(68) The fluid supply headers **38** and **39** are configured to supply the supercritical fluid into the main body **31**. Further, the fluid discharge header **40** is configured to guide and discharge the supercritical fluid within the main body **31** to the outside of the main body **31**. Further, the supercritical fluid discharged to the outside of the main body **31** through the fluid discharge header **40** contains the IPA liquid dissolved in the supercritical fluid from the surface of the wafer **W**.

(69) Within this drying unit **18**, the IPA liquid between patterns formed on the wafer **W** is slowly dissolved in the supercritical fluid as it comes into contact with the supercritical fluid in a high pressure state (e.g., 16 MPa), and the IPA liquid between the patterns is gradually replaced by the

supercritical fluid. Finally, gaps between the patterns are filled with only the supercritical fluid.

(70) By decompressing the main body **31** to an atmospheric pressure from the high pressure state after the IPA liquid between the patterns is removed, the CO.sub.2 is changed from the supercritical state to a gaseous state, so that the gaps between the patterns are occupied by only the gas. In this way, the IPA liquid between the patterns is removed and the drying processing of the wafer W is completed.

(71) Here, the supercritical fluid has a smaller viscosity than a liquid (e.g., the IPA liquid) and has a higher ability to dissolve the liquid. Further, there is no interface between the supercritical fluid and the liquid or the gas in an equilibrium state with the supercritical fluid. Accordingly, in the drying processing with the supercritical fluid, the liquid can be dried without being affected by a surface tension. Therefore, according to the present exemplary embodiment, it is possible to suppress the occurrence of the pattern collapse during the drying processing.

(72) A processing time of at least the drying processing is different between the liquid processing unit **17** configured to process the wafer WA and the liquid processing unit **17** configured to process the wafer WB.

(73) Further, although it has been described in the present exemplary embodiment that the IPA liquid is used as the anti-drying liquid and CO.sub.2 in the supercritical state is used as the processing fluid, a liquid other than the IPA may be used as the anti-drying liquid and a fluid other than CO.sub.2 in the supercritical state may be used as the processing fluid.

(74) <Substrate Processing Flow>

(75) Hereinafter, a processing flow of the wafer W in the substrate processing system **1** will be described with reference to FIG. **6**. FIG. **6** is a flowchart illustrating a sequence of a series of substrate processings performed in the substrate processing system **1** according to the exemplary embodiment. The series of substrate processings shown in FIG. **6** are performed under the control of the controller **61**.

(76) Also, herein, a sequence of a series of substrate processings performed on a single wafer W is described as an example. In the substrate processing system **1**, the series of substrate processings shown in FIG. **6** are performed in parallel on the plurality of wafers WA and the plurality of wafers WB.

(77) In the substrate processing system **1**, the transfer device **13** takes out a wafer W from the carrier C and places the wafer W on the transit unit **14** (process S101). Specifically, the transfer device **13** takes out a wafer WA from the carrier CA by using the first holder **131A** and places the taken wafer WA on the transit unit **14A**. Also, the transfer device **13** takes out a wafer WB from the carrier CB by using the second holder **131B** and places the taken wafer WB on the transit unit **14B**.

(78) Then, in the substrate processing system **1**, there is performed a first transfer processing (process S102). In the first transfer processing, the transfer device **16** takes out the wafer W from the transit unit **14** and transfers the wafer W to the liquid processing unit **17**. Specifically, the transfer device **16** takes out the wafer WA from the transit unit **14A** by using the first holder **161A** and transfers the taken wafer WA to the liquid processing unit **17** of the processing block 5A. Also, the transfer device **16** takes out the wafer WB from the transit unit **14B** by using the second holder **161B** and transfers the taken wafer WB to the liquid processing unit **17** of the processing block 5B.

(79) Thereafter, in the substrate processing system **1**, the liquid processing unit **17** performs the liquid processing (process S103). Specifically, the liquid processing unit **17** removes the particle and the natural oxide film from the top surface of the wafer W, for example, by supplying various processing liquids onto the top surface of the wafer W which is the pattern formation surface. Then, the liquid processing unit **17** forms the liquid film of the IPA liquid on the top surface of the wafer W, for example, by supplying the IPA liquid onto the top surface of the wafer W after being subjected to the cleaning processing.

(80) Subsequently, in the substrate processing system **1**, there is performed a second transfer processing (process S104). In the second transfer processing, the transfer device **16** takes out the

wafer W having the liquid film formed thereon from the liquid processing unit **17** and then transfers the wafer W to the drying unit **18**. Specifically, the transfer device **16** takes out the wafer WA from the liquid processing unit **17** by using the first holder **161A** and transfers the taken wafer WA to the drying unit **18** of the processing block 5A. Also, the transfer device **16** takes out the wafer WB from the liquid processing unit **17** by using the second holder **161B** and transfers the taken wafer WB to the drying unit **18** of the processing block 5B.

(81) Thereafter, in the substrate processing system **1**, the drying unit **18** performs the drying processing (process S**105**). In the drying processing, the drying unit **18** dries the wafer W having the liquid film formed thereon by bringing the wafer W into contact with the supercritical fluid.

(82) Then, in the substrate processing system **1**, there is performed a third transfer processing (process S**106**). In the third transfer processing, the transfer device **16** takes the wafer W after being subjected to the drying processing from the drying unit **18** and transfers the wafer W to the transit unit **14**. Specifically, the transfer device **16** takes out the wafer WA from the drying unit **18** by using the first holder **161A** and places the taken wafer WA on the transit unit **14A**. Also, the transfer device **16** takes out the wafer WB from the drying unit **18** by using the second holder **161B** and places the taken wafer WB on the transit unit **14B**.

(83) Subsequently, in the substrate processing system **1**, the transfer device **13** takes out the wafer W from the transit unit **14** and carries the wafer W into the carrier C (process S**108**). Specifically, the transfer device **13** takes out the wafer WA from the transit unit **14A** by using the first holder **131A** and places the taken wafer WA in the carrier CA. Also, the transfer device **13** takes out the wafer WB from the transit unit **14B** by using the second holder **131B** and places the taken wafer WB in the carrier CB. Upon the completion of this carry-out processing, the series of substrate processings on the single wafer W are ended.

(84) As described above, in the substrate processing system **1**, the series of substrate processings are performed in parallel on the plurality of wafers WA and the plurality of wafers WB. Processing times of at least the liquid processing (process S**103**) and the drying processing (process S**105**) are different between the series of substrate processings (process S**101** to process S**107**) performed on the wafer WA and the series of substrate processings (process S**101** to process S**107**) performed on the wafer WB.

(85) Herein, if another wafer W is being transferred by the transfer device **16** at a timing when the second transfer processing of the wafer W after being subjected to the liquid processing is to be performed, the wafer W having the liquid film formed thereon is required to stand by until the transfer of the another wafer W is completed.

(86) Further, if a state of the liquid film on the surface of the wafer W is changed as the liquid film is dried or the like during the standby period, the problem such as the collapse of the pattern formed on the wafer W may occur in the subsequent drying processing, which may cause the reduction in yield of wafers W. Such an overlap of wafer transfer timings is likely to occur particularly when a plurality of recipes having different processing times is performed in parallel as in the substrate processing system **1** according to the exemplary embodiment.

(87) Therefore, in the present exemplary embodiment, a start timing of the first transfer processing of a subsequent wafer W is determined such that a time period of the second transfer processing of a preceding wafer W (e.g., the wafer WA) does not overlap with a time period of the second transfer processing of a subsequent wafer W (e.g., the wafer WB).

(88) Accordingly, it is possible to suppress the transfer of the wafer WB by the transfer device **16** at a timing when the wafer WA after being subjected to the liquid processing is transferred. In other words, it is possible to suppress the occurrence of standby time of the wafer WA after being subjected to the liquid processing. Therefore, the substrate processing system **1** according to the exemplary embodiment can suppress the non-uniformity in transfer time of the wafer having the liquid film on its surface, i.e., the time period from the completion of the liquid processing to the carry-in of the wafer into the drying unit **18**.

(89) <Detail of Subsequent Recipe Start Timing Adjustment Processing>

(90) This will be described in detail with reference to FIG. 7. FIG. 7 is a timing chart for explaining a subsequent recipe start timing adjustment processing according to the exemplary embodiment.

The processings shown in FIG. 7 are performed under the control of the controller 61.

(91) Hereinafter, the series of substrate processings performed on the wafer WA will be described as “recipe A” and the series of substrate processings performed on the wafer WB will be described as “recipe B”. FIG. 7 illustrates an example of a timing chart when the recipe A and the recipe B are performed in parallel.

(92) The timing chart for the recipe A in FIG. 7 shows an example where a plurality of wafers WA1 and WA2 is processed in this order in a pair of a liquid processing unit 17A1 and a drying unit 18A1 placed in the processing block 5A. Also, the timing chart for the recipe B in FIG. 7 shows an example where a plurality of wafers WB1 and WB2 is processed in this order in a pair of a liquid processing unit 17B1 and a drying unit 18B1 placed in the processing block 5B.

(93) Further, FIG. 7 illustrates a processing period of the liquid processing on the wafer WA1 as “WA1” and a processing period of the liquid processing on the wafer WA2 as “WA2” in the liquid processing performed in the liquid processing unit 17A1. Also, FIG. 7 illustrates a processing period of the drying processing on the wafer WA1 as “WA1” and a processing period of the drying processing on the wafer WA2 as “WA2” in the liquid processing performed in the drying unit 18A1. The same is applied to the liquid processing unit 17B1 and the drying unit 18B1, and a processing period of the processing on the wafer WB1 is illustrated as “WB1” and a processing period of the processing on the wafer WB2 is illustrated as “WB2”.

(94) Further, FIG. 7 illustrates a processing period of the first transfer processing as “S1”, a processing period of the second transfer processing as “S2” and a processing period of the third transfer processing as “S3”.

(95) For example, it is assumed that when the liquid processing is performed on the wafer WA1 in the liquid processing unit 17A1, the recipe B for the wafer WB1 can be started. In this case, if the recipe B for the wafer WB1 is started immediately, the second transfer processing of the wafer WA1 and the second transfer processing of the wafer WB1 may overlap with each other depending on the timing when the recipe B can be started.

(96) Therefore, after the first transfer processing of the preceding wafer WA1 is started and before the first transfer processing of the subsequent wafer WB1 is started, the controller 61 determines a start timing of the first transfer processing of the wafer WB1 in the following order.

(97) First, the controller 61 specifies an expected performance period of the second transfer processing of the wafer WA1 based on the processing times of the first transfer processing, the liquid processing and the second transfer processing of the wafer WA1. Specifically, all the processing times of the first transfer processing, the liquid processing and the second transfer processing of the wafer WA1 are already known. Further, the controller 61 may specify a time point at which the processing times of the first transfer processing and the liquid processing have elapsed from a start timing t1 of the first transfer processing of the wafer WA1 as an expected start timing t2 of the second transfer processing of the wafer WA1. Furthermore, the controller 61 may specify a time period in which the processing time of the second transfer processing has elapsed from the expected start timing t2 as the expected performance period of the second transfer processing of the wafer WA1.

(98) Then, the controller 61 specifies an expected performance period of the second transfer processing of the wafer WB1 when it is assumed that the first transfer processing of the wafer WB1 is started, based on the processing times of the first transfer processing, the liquid processing and the second transfer processing of the wafer WB1. Specifically, all the processing times of the first transfer processing, the liquid processing and the second transfer processing of the wafer WB1 are already known. The controller 61 may specify a time point at which the processing times of the first transfer processing and the liquid processing have elapsed from, for example, a timing when

the recipe B for the wafer WB1 can be started as an expected start timing t3 of the second transfer processing of the wafer WB1. Further, the controller **61** may specify a time period in which the processing time of the second transfer processing has elapsed from the expected start timing t3 as the expected performance period of the second transfer processing of the wafer WB1.

(99) Then, the controller **61** determines whether an interval between the expected start timing t2 of the second transfer processing of the wafer WA1 and the expected start timing t3 of the second transfer processing of the wafer WB1 is shorter than a predetermined interval time I. The interval time I is set longer than at least the processing time of the second transfer processing.

(100) If the interval is shorter than the interval time I, the controller **61** determines a timing when the interval is equal to or longer than the interval time I as a start timing t4 of the first transfer processing of the wafer WB1. Specifically, the controller **61** determines the start timing t4 of the first transfer processing of the wafer WB1 such that the interval becomes equal to the interval time I.

(101) As described above, in the substrate processing system **1** according to the exemplary embodiment, the start timing of the first transfer processing of the wafer WB1 is determined such that the time period of the second transfer processing of the preceding wafer WA1 does not overlap with the time period of the second transfer processing of the subsequent wafer WB1.

(102) Specifically, the first transfer processing of the wafer WB1 is started at a timing when an interval between the expected start timing t2 of the second transfer processing of the wafer WA1 and the expected start timing t3 of the second transfer processing of the wafer WB1 is equal to or longer than the interval time I.

(103) Accordingly, it is possible to suppress the transfer of the wafer WB1 by the transfer device **16** at a timing when the second transfer processing of the wafer WA1 after being subjected to the liquid processing is performed. In other words, it is possible to suppress the occurrence of standby time of the wafer WA after being subjected to the liquid processing. Therefore, the substrate processing system **1** according to the exemplary embodiment can suppress the non-uniformity in transfer time of the wafer having the liquid film on its surface.

(104) When the second transfer processing of the wafer WA1 is completed, the recipe A for the subsequent wafer WA2 can be performed. Here, the controller **61** specifies an expected start timing t5 of the second transfer processing of the wafer WA2 when it is assumed that the recipe A for the wafer WA2 is started immediately. Further, the controller **61** determines whether an interval between the expected start timing t3 of the second transfer processing of the preceding wafer WB1 and the expected start timing t5 of the second transfer processing of the subsequent wafer WA2 is shorter than the interval time I. In the illustrated example, the interval between the expected start timing t3 and the expected start timing t5 is equal to or longer than the interval time I. Therefore, the controller **61** starts the first transfer processing of the wafer WA2 at a timing when the recipe A for the wafer WA2 can be performed, i.e., when the second transfer processing of the wafer WA1 is completed.

(105) The controller **61** also determines a start timing t7 of the first transfer processing of the wafer WB2 in the same procedure as described above. That is, the controller **61** determines the start timing t7 of the first transfer processing of the wafer WB2 such that an interval between the expected start timing t5 of the second transfer processing of the wafer WA2 and an expected start timing t6 of the second transfer processing of the wafer WB2 becomes equal to the interval time I.

(106) In the above-described example, it is determined whether the interval between the expected start timing of the second transfer processing of the preceding wafer W and the expected start timing of the second transfer processing of the subsequent wafer W is shorter than the interval time I. However, the present disclosure is not limited thereto. For example, the controller **61** may determine whether an interval between an expected end timing of the second transfer processing of the preceding wafer W and the expected start timing of the second transfer processing of the subsequent wafer W is shorter than the interval time I. That is, the controller **61** may determine

whether an interval between the expected performance period of the second transfer processing of the preceding wafer W and the expected performance period of the second transfer processing of the subsequent wafer W is shorter than the interval time I. In this case, the interval time I may be at least 0, and may be shorter than the interval time I which is set for the interval between the expected start timings.

(107) Further, the interval between the expected start timing of the second transfer processing of the preceding wafer W and the expected start timing of the second transfer processing of the subsequent wafer W can be set equal to or longer than the interval time by the following two methods. A first method is to advance the expected start timing of the second transfer processing of the subsequent wafer W, and a second method is to delay the expected start timing of the second transfer processing of the subsequent wafer W. In this case, the controller **61** may adopt the method of delaying rather than the method of advancing. Accordingly, it is possible to suppress a processing of the subsequent wafer W from overtaking a processing of the preceding wafer W. Therefore, it is possible to suppress, for example, complexity of the system.

(108) <Transfer Prohibition Period>

(109) Before the expected start timing of the second transfer processing of the preceding wafer W, the controller **61** may set a transfer prohibition period ending at the expected start timing. This will be described with reference to FIG. **8**. FIG. **8** is a timing chart for explaining an example of a subsequent recipe start timing adjustment processing in consideration of a transfer prohibition period. The processings shown in FIG. **8** are performed under the control of the controller **61**.

(110) The upper chart in FIG. **8** shows an example where the start timing t13 of the first transfer processing of the subsequent wafer WB1 is included in a transfer prohibition period P. The lower chart in FIG. **8** shows an example where the start timing t13 is delayed such that the start timing t13 of the first transfer processing of the subsequent wafer WB1 is not included in the transfer prohibition period P.

(111) As shown in FIG. **8**, before an expected start timing t11 of the second transfer processing of the preceding wafer WA2, the transfer prohibition period P ending at the expected start timing t11 is set. First, the controller **61** temporarily determines a start timing t13 of the first transfer processing of the wafer WB1. The start timing t13 is temporarily determined such that an interval between the expected start timing t11 of the second transfer processing of the preceding wafer WA2 and an expected start timing t12 of the second transfer processing of the subsequent wafer WB1 becomes equal to the interval time I. Then, the controller **61** determines whether the temporarily determined start timing t13 is included in the transfer prohibition period P.

(112) If the temporarily determined start timing t13 is included in the transfer prohibition period P as shown in the upper chart in FIG. **8**, the controller **61** delays the temporarily determined start timing t13 such that the start timing t13 does not overlap with the transfer prohibition period P as shown in the lower chart in FIG. **8**. Specifically, the controller **61** starts the first transfer processing of the wafer WB1 at a timing when the interval between the expected start timings t11 and t12 is equal to or longer than the interval time I and the start timing t13 does not overlap with the transfer prohibition period P. Herein, the first transfer processing of the wafer WB1 cannot be started until the second transfer processing of the wafer WA2 is ended. Therefore, the first transfer processing of the wafer WB1 is started at a timing when the second transfer processing of the wafer WA2 is ended.

(113) As described above, the transfer prohibition period P may be set right before the expected start timing of the second transfer processing of the preceding wafer W, and a transfer of another wafer W may be prohibited in the transfer prohibition period P. Accordingly, the second transfer processing of the preceding wafer W can be started more reliably at the expected start timing. Also, if the preceding wafer WA1 is present in the drying unit **18A1** as the carry-in destination of the wafer WA2, the subsequent wafer WA2 cannot be carried into the drying unit **18A1**. For this reason, the processing of carrying the preceding wafer WA1 out of the drying unit **18B1**, i.e., the

third transfer processing of the wafer WA1, is allowed to be performed in the transfer prohibition period P.

(114) <Method of Determining Start Timing of First Transfer Processing in Same Recipe>

(115) In the above-described example, the start timing of the first transfer processing between the different recipes A and B is determined based on the interval time I. However, the present disclosure is not limited thereto. A start timing of the first transfer processing in the same recipe may also be determined based on the interval time I. This will be described in detail with reference to FIG. 9. FIG. 9 is a timing chart for explaining an example of a method of determining a start timing of a first transfer processing in the same recipe. The processings shown in FIG. 9 are performed under the control of the controller 61.

(116) FIG. 9 shows an example where a plurality of wafers WA1 and WA3 is processed in this order in a pair of the liquid processing unit 17A1 and the drying unit 18A1 placed in the processing block 5A. Also, FIG. 9 shows an example where a plurality of wafers WA2 and WA4 is processed in this order in a pair of a liquid processing unit 17A2 and a drying unit 18A2 placed in the processing block 5A.

(117) As shown in FIG. 9, the controller 61 specifies an expected start timing t21 of the second transfer processing of the preceding wafer WA1 and an expected start timing t22 of the second transfer processing of the subsequent wafer WA2. Further, the controller 61 determines a start timing t23 of the first transfer processing of the wafer WA2 such that an interval between the expected start timings t21 and t22 becomes equal to the interval time I.

(118) As described above, the controller 61 may determine the start timing of the first transfer processing of the subsequent wafer WA with respect to the wafers WA processed according to the same recipe (herein, the recipe A) such that the interval between expected start timings of the second transfer processing becomes equal to or longer than the interval time. The same is applied between the wafers WA2 and WA3 and between the wafers WA3 and WA4.

(119) Also, the controller 61 may determine a start timing t24 of the first transfer processing of the subsequent wafer WA3 with respect to the wafers WA1 and WA3 processed in the pair of the liquid processing unit 17A1 and the drying unit 18A1 in the following order.

(120) First, the controller 61 specifies an expected end timing t25 of the third transfer processing of the preceding wafer WA1 based on the already-known processing time of the third transfer processing. Then, the controller 61 determines the start timing t24 of the first transfer processing of the subsequent wafer WA3 such that the second transfer processing of the wafer WA3 is started at the expected end timing t25. In other words, the controller 61 determines the start timing t24 of the first transfer processing of the subsequent wafer WA3 such that the end timing of the third transfer processing of the preceding wafer WA1 is the same as the start timing of the second transfer processing of the subsequent wafer WA3.

(121) Accordingly, the start of the second transfer processing of the subsequent wafer WA3 is delayed, for example, until the third transfer processing of the preceding wafer WA1 is ended. Therefore, it is possible to suppress the non-uniformity in processing time of the second transfer processing of the wafer WA3. Also, since the end timing of the third transfer processing of the preceding wafer WA1 is set to be same as the start timing of the second transfer processing of the subsequent wafer WA3, it is possible to efficiently transfer the wafers WA1 and WA3.

(122) However, the end timing of the third transfer processing of the preceding wafer WA1 is not necessarily same as the start timing of the second transfer processing of the subsequent wafer WA3. That is, the controller 61 may determine the start timing t24 of the first transfer processing of the wafer WA3 such that the liquid processing of the subsequent wafer WA3 is ended within a predetermined period of time after the third transfer processing of the preceding wafer WA1 is ended.

(123) <Method of Determining Priority when Two Recipes can be Started>

(124) When both the recipe A and the recipe B can be started, the controller 61 may priorly perform

a recipe other than the recipe performed last between the recipe A and the recipe B.

(125) This will be described in detail with reference to FIG. 10. FIG. 10 is a timing chart for explaining an example of a method of determining a priority when two recipes can be started. The processings shown in FIG. 10 are performed under the control of the controller 61.

(126) FIG. 10 shows an example where the WA1 is processed in the pair of the liquid processing unit 17A1 and the drying unit 18A1 placed in the processing block 5A. Also, FIG. 10 shows an example where the wafer WA2 is processed in the pair of the liquid processing unit 17A2 and the drying unit 18A2 placed in the processing block 5A. Further, FIG. 10 shows an example where the wafer WB1 is processed in the pair of the liquid processing unit 17B1 and the drying unit 18B1 placed in the processing block 5B.

(127) As shown in FIG. 10, it is assumed that both the wafer WA2 processed according to the recipe A and the wafer WB1 processed according to the recipe B can be selected as a wafer W subsequent to the preceding wafer WA1. In this case, the preceding wafer WA1 is the wafer WA processed according to the recipe A, and, thus, the controller 61 priorly starts a processing of the wafer WB1 processed according to the recipe B as the subsequent wafer W. Specifically, the controller 61 first starts the first transfer processing of the wafer WB1 at a start timing t31, and then, starts the first transfer processing of the wafer WA2 at a start timing t32. The start timing t32 is also an end timing of the first transfer processing of the wafer WB1.

(128) As described above, when both the recipe A and the recipe B can be started, the controller 61 may priorly start performing a recipe other than the recipe performed last between the recipe A and the recipe B. Accordingly, it is possible to suppress the difference in progress between the recipe A and the recipe B.

(129) <Transfer Time>

(130) Processing times of the first transfer processing, the second transfer processing and the third transfer processing (hereinafter, referred to as “transfer times”) may be previously fixed. This will be described in detail with reference to FIG. 11 and FIG. 12. FIG. 11 illustrates an example of transfer time information. Also, FIG. 12 is a flowchart illustrating an example of a sequence of a third transfer processing. The processings shown in FIG. 12 are performed under the control of the controller 61.

(131) The storage 62 of the control device 6 previously stores therein, for example, transfer time information shown in FIG. 11. As shown in FIG. 11, the transfer time information is information in which the time (operation time) from the start to the end of an operation is associated with each individual operation for transfer.

(132) Specifically, each of the first transfer processing, the second transfer processing and the third transfer processing includes “unit-front movement”, “wafer receipt” and “wafer carry-out” as individual operations. The “unit-front movement” is an operation of moving the transfer device 16 from a certain movement source to a certain movement destination. The “wafer receipt” is an operation of receiving the wafer from the movement destination. The “wafer carry-out” is an operation of delivering the wafer to the movement destination.

(133) For example, according to the transfer time information shown in FIG. 11, the operation time of the unit-front movement from the liquid processing unit 17 to the drying unit 18 is defined as “a1” in case that wafer holding is “absence” and “a3” in case that the wafer holding is “presence”. The operation time “a3” is set longer than the operation time “a1”. This is to ensure that the moving speed in a state where the wafer W is held is lower than the moving speed in a state where the wafer W is not held from the viewpoint of suppressing the misalignment of the wafer W. Likewise, according to the transfer time information shown in FIG. 11, the operation time of the unit-front movement from the drying unit 18 to the transit unit 14 is defined as “a2” in case that the wafer holding is “absence” and “a4” in case that the wafer holding is “presence”.

(134) Further, according to the transfer time information shown in FIG. 11, the operation time of the “wafer receipt” of receiving the wafer W from the drying unit 18 is defined as “a5”, and the

operation time of the “wafer carry-out” of placing the wafer W on the transit unit **14** is defined as “a6”.

(135) Hereinafter, an example where the third transfer processing is performed according to the transfer time information will be described with reference to FIG. **12**. As shown in FIG. **12**, the controller **61** performs the unit-front movement from the liquid treatment unit **17** to the drying unit **18** (process S201), and then, performs the wafer receipt from the drying unit **18** (process S202). Thereafter, the controller **61** performs the unit-front movement from the drying unit **18** to the transit unit **14** (process S203), and then, performs the wafer carry-out to the transit unit **14** (process S204).

(136) In the process S201, the controller **61** first actuates a timer for the operation time “a1” (process S201a). Subsequently, the controller **61** moves the transfer device **16** to the front of the drying unit **18** (process S201b). Then, the controller **61** makes the transfer device **16** stand by in front of the drying unit **18** until the timer for “a1” actuated in the process S201a is timed out (process S201c).

(137) When the timer for “a1” is timed out in the process S201c, the controller **61** starts the process S202. In the process S202, the controller **61** first actuates a timer for the operation time “a5” (process S202a). Subsequently, the controller **61** receives the wafer W from the drying unit **18** (process S202b). Then, the controller **61** makes the transfer device **16** stand by in front of the drying unit **18** until the timer for “a5” actuated in the process S202a is timed out (process S202c).

(138) When the timer for “a5” is timed out in the process S202c, the controller **61** starts the process S203. In the process S203, the controller **61** first actuates a timer for the operation time “a4” (process S203a). Subsequently, the controller **61** moves the transfer device **16** to the transit unit **14** (process S203b). Then, the controller **61** makes the transfer device **16** stand by in front of the transit unit **14** until the timer for “a4” actuated in the process S203a is timed out (process S203c).

(139) When the timer for “a4” is timed out in the process S203c, the controller **61** starts the process S204. In the process S204, the controller **61** first actuates a timer for the operation time “a6” (process S204a). Subsequently, the controller **61** places the wafer W on the transit unit **14** (process S204b). Then, the controller **61** makes the transfer device **16** stand by in front of the transit unit **14** until the timer for “a6” actuated in the process S204a is timed out (process S204c).

(140) Herein, the third transfer processing has been described as an example, but the same is applied to the first transfer processing and the second transfer processing. That is, in the first transfer processing, the controller **61** moves the transfer device **16** from the transit unit **14** to the liquid processing unit **17**, and then, makes the transfer device **16** stand by until a predetermined period of time elapses from the start of the first transfer processing. Further, in the second transfer processing, the controller **61** moves the transfer device **16** from the liquid processing unit **17** to the drying unit **18**, and then, makes the transfer device **16** stand by until a predetermined period of time elapses from the start of the second transfer processing.

(141) By controlling as described above, the transfer times of the first to third transfer processings can be fixed. In other words, it is possible to suppress an increase or a decrease in transfer time of the first to third transfer processings. Accordingly, it is possible to suppress a fluctuation in a transfer cycle of the wafers W and thus possible to stably transfer the wafers W on a predetermine cycle.

(142) As described above, a substrate processing apparatus (for example, substrate processing system **1**) includes a transit unit (for example, transit unit **14**); multiple liquid processing units (for example, liquid processing units **17**); multiple drying units (for example, drying units **18**); a transfer device (for example, transfer device **16**) and a controller (for example, controller **61**). The transit unit is configured to accommodate therein multiple substrates (for example, wafer **17**). The multiple liquid processing units are configured to form a liquid film on a surface of the substrate. The multiple drying units are configured to dry the substrate by bringing the substrate having the liquid film formed on the surface thereof into contact with a supercritical fluid. The transfer device

is configured to transfer the substrate. The controller is configured to control the transit unit, the multiple liquid processing units, the multiple drying units and the transfer device to perform two recipes, which includes a first recipe (for example, one of the recipe A or the recipe B) and a second recipe (for example, the other of the recipe A or the recipe B), in parallel, each of the first recipe and the second recipe including a first transfer processing of transferring the substrate from the transit unit to the liquid processing unit, a liquid processing performed by the liquid processing unit, a second transfer processing of transferring the substrate from the liquid processing unit to the drying unit and a drying processing performed by the drying unit, the first recipe and the second recipe having different processing times of at least the liquid processing and the drying processing. Further, the controller determines, while the first recipe on a first substrate in the multiple substrates is being performed, when the second recipe on a second substrate in the multiple substrates is started, a start timing of the first transfer processing of the second substrate such that a time period of the second transfer processing of the first substrate does not overlap with a time period of the second transfer processing of the second substrate.

(143) Further, when an interval between an expected start timing of the second transfer processing of the second substrate when it is assumed that the first transfer processing of the second substrate is started and a start timing of the second transfer processing of the first substrate is shorter than a predetermined interval time (for example, interval time I), the controller starts the first transfer processing of the second substrate at a timing when the interval becomes equal to or longer than the interval time.

(144) Therefore, in the substrate processing apparatus according to the exemplary embodiment, the non-uniformity in transfer time of the wafer having the liquid film on its surface can be suppressed even when the plurality of recipes having different processing times is performed in parallel.

(145) The controller may start the first transfer processing of the second substrate at the timing when the interval becomes equal to or longer than the interval time such that the second transfer processing of the second substrate is started after the second transfer processing of the first substrate is ended.

(146) Accordingly, it is possible to suppress the processing of the subsequent substrate from overtaking the processing of the preceding substrate. Therefore, it is possible to suppress, for example, the complexity of the system.

(147) When the start timing of the first transfer processing is included in a transfer prohibition period ending at a start timing of the second transfer processing of another one of the multiple substrates assuming that the first transfer processing of the second substrate is started at the timing when the interval becomes equal to or longer than the interval time, the controller may start the first transfer processing of the second substrate after the second transfer processing of the another one of the multiple substrates is ended.

(148) Accordingly, the second transfer processing of the preceding substrate can be started more reliably at the expected start timing.

(149) The recipe may include a third transfer processing of transferring the substrate from the drying unit to the transit unit. In this case, when the first recipe on a third substrate in the multiple substrates is started during the drying processing of the first substrate, the controller may start the first transfer processing of the third substrate such that the liquid processing of the third substrate is ended within a predetermined period of time after the third transfer processing of the first substrate is ended.

(150) Accordingly, the start of the second transfer processing of the subsequent substrate is delayed, for example, until the third transfer processing of the preceding substrate is ended. Therefore, it is possible to suppress the non-uniformity in processing time of the second transfer processing of the subsequent substrate.

(151) When both the first recipe and the second recipe are allowed to be started, the controller may start performing the recipe other than the recipe performed last between the first recipe and the

second recipe.

(152) Accordingly, it is possible to suppress the difference in progress between the recipe A and the recipe B.

(153) In the first transfer processing, the controller may move the transfer device from the transit unit to the liquid processing unit, and then, make the transfer device stand by until a predetermined period of time elapses from a time when the first transfer processing is started. Further, in the second transfer processing, the controller may move the transfer device from the liquid processing unit to the drying unit, and then, make the transfer device stand by until a predetermined period of time elapses from a time when the second transfer processing is started.

(154) Accordingly, it is possible to suppress the increase or the decrease in transfer time of the first to third transfer processings. Accordingly, it is possible to suppress the fluctuation in the transfer cycle of the substrates, and thus, possible to stably transfer the substrates on the predetermine cycle.

(155) While the present disclosure has been described with reference to the exemplary embodiments, the present disclosure is not limited to the exemplary embodiments but may be variously modified without departing from the spirit thereof. For example, in the above-described exemplary embodiment, the substrate processing system **1** is provided with one transfer device **16**. However, the number of transfer devices **16** is not limited to one. A plurality of transfer devices **16** may be provided as long as they are provided in common for the liquid processing unit **17** and drying unit **18** in each of the plurality of pairs.

(156) The exemplary embodiments disclosed herein are illustrative in all aspects and do not limit the present disclosure. In fact, the above exemplary embodiments can be embodied in various forms. Further, the above-described exemplary embodiments may be omitted, replaced and modified in various ways without departing from the scope and the spirit of the appended claims.

(157) According to the present disclosure, it is possible to suppress non-uniformity in transfer time of a wafer having a liquid film on its surface even when a plurality of recipes having different processing times is performed in parallel.

Claims

1. A substrate processing apparatus, comprising: a transit unit configured to accommodate therein multiple substrates; multiple liquid processing units each configured to form a liquid film on a surface of the substrate; multiple drying units each configured to dry the substrate by bringing the substrate having the liquid film formed on the surface thereof into contact with a supercritical fluid; a transfer device configured to transfer the substrate; and a controller configured to control the transit unit, the multiple liquid processing units, the multiple drying units and the transfer device to perform two recipes, which includes a first recipe and a second recipe, in parallel, each of the first recipe and the second recipe including a first transfer processing of transferring the substrate from the transit unit to the liquid processing unit, a liquid processing performed by the liquid processing unit, a second transfer processing of transferring the substrate from the liquid processing unit to the drying unit and a drying processing performed by the drying unit, the first recipe and the second recipe having different processing times of at least the liquid processing and the drying processing, wherein while the first recipe on a first substrate in the multiple substrates is being performed, when the second recipe on a second substrate in the multiple substrates is started, the controller determines a start timing of the first transfer processing of the second substrate such that a time period of the second transfer processing of the first substrate does not overlap with a time period of the second transfer processing of the second substrate.

2. The substrate processing apparatus of claim 1, wherein when an interval between an expected start timing of the second transfer processing of the second substrate when it is assumed that the first transfer processing of the second substrate is started and a start timing of the second transfer

- processing of the first substrate is shorter than a predetermined interval time, the controller starts the first transfer processing of the second substrate at a timing when the interval becomes equal to or longer than the interval time.
3. The substrate processing apparatus of claim 2, wherein the controller starts the first transfer processing of the second substrate at the timing when the interval becomes equal to or longer than the interval time such that the second transfer processing of the second substrate is started after the second transfer processing of the first substrate is ended.
4. The substrate processing apparatus of claim 2, wherein when the start timing of the first transfer processing of the second substrate is included in a transfer prohibition period ending at a start timing of the second transfer processing of another one of the multiple substrates assuming that the first transfer processing of the second substrate is started at the timing when the interval becomes equal to or longer than the interval time, the controller starts the first transfer processing of the second substrate after the second transfer processing of the another one of the multiple substrates is ended.
5. The substrate processing apparatus of claim 1, wherein the recipe includes a third transfer processing of transferring the substrate from the drying unit to the transit unit, and when the first recipe on a third substrate in the multiple substrates is started during the drying processing of the first substrate, the controller starts the first transfer processing of the third substrate such that the liquid processing of the third substrate is ended within a predetermined period of time after the third transfer processing of the first substrate is ended.
6. The substrate processing apparatus of claim 1, wherein when both the first recipe and the second recipe are allowed to be started, the controller starts performing the recipe other than the recipe performed last between the first recipe and the second recipe.
7. The substrate processing apparatus of claim 1, wherein, in the first transfer processing, the controller moves the transfer device from the transit unit to the liquid processing unit, and then, makes the transfer device stand by until a predetermined period of time elapses from a time when the first transfer processing is started.
8. The substrate processing apparatus of claim 1, wherein, in the second transfer processing, the controller moves the transfer device from the liquid processing unit to the drying unit, and then, makes the transfer device stand by until a predetermined period of time elapses from a time when the second transfer processing is started.
9. A substrate processing method, comprising: performing two recipes, which includes a first recipe and a second recipe, in parallel, each of the first recipe and the second recipe including a first transfer processing of transferring a substrate of multiple substrates from a transit unit configured to accommodate therein the multiple substrates to any one of multiple liquid processing units each configured to form a liquid film on a surface of the substrate by using a transfer device; a liquid processing performed by the liquid processing unit; a second transfer processing of transferring the substrate from the liquid processing unit to any one of multiple drying units each configured to dry the substrate by bringing the substrate into contact with a supercritical fluid by using the transfer device; and a drying processing performed by the drying unit, the first recipe and the second recipe having different processing times of at least the liquid processing and the drying processing, wherein the performing of the two recipes in parallel includes: while the first recipe on a first substrate in the multiple substrates is being performed, when the second recipe on a second substrate in the multiple substrates is started, determining a start timing of the first transfer processing of the second substrate such that a time period of the second transfer processing of the first substrate does not overlap with a time period of the second transfer processing of the second substrate; and starting the first transfer processing of the second substrate at the start timing determined in the determining of the start timing.
-